CASCADE

’26

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PRESENTS

THEMES & PROBLEM STATEMENTS

24-Hour Offline Hackathon · IIIT Nagpur · TantraFiesta 2026

25–26 October 2026 · Four Themes · Twelve Problems

PRIZE POOL : 2,00,000 INR

THEMES AT A GLANCE

Teams select one preferred theme during registration. Each theme has three open-ended systems challenges, many with an adversarial “red team” twist.

1 THE FORGOTTEN POPULATION

Design specifically for groups mainstream technology does not optimize for.

- 1.1 The City That Sleeps While You Work
- 1.2 The Invisible Worker
- 1.3 The Human-Powered Emergency Network

2 CRISIS BEFORE CRISIS

Predict the conditions that create disasters instead of only responding after they happen.

- 2.1 The First 10 Minutes
- 2.2 The 72-Hour Warning
- 2.3 The Critical 1%: Hidden Community Vulnerability

3 ATTENTION ECONOMY

Build technology that protects human attention instead of extracting it.

- 3.1 The Attention Firewall
- 3.2 The Infinite Scroll Intervention Engine
- 3.3 The Attention Digital Twin

4 INFORMATION ASYMMETRY

Give ordinary people the information that institutions already have, and make it impossible for that information to be hidden again.

- 4.1 The Governance Illusion: DAO Accountability
- 4.2 The Black Hole Ledger: Verifiable NGO Fund Flows
- 4.3 The Identity Monopoly: Decentralized Portable Proofs

THEME 1

THE FORGOTTEN POPULATION

Core idea: Design specifically for groups mainstream technology does not optimize for.

THEME 1 • PROBLEM 1

THE CITY THAT SLEEPS WHILE YOU WORK

CONTEXT

A city is designed around people who work during the day, but millions of workers (security guards, nurses, factory workers, sanitation workers, delivery workers, call-center employees) live on a completely different schedule. At 2:30 AM, a worker may need food, medicine, transportation, emergency assistance, or a safe route home. Existing navigation and service-discovery systems treat nighttime as just a different timestamp.

CHALLENGE

Build a Night-Time Decision Engine that understands the changing conditions of a city throughout the night. It should consider:

- Availability of public/private transportation
- Operating hours of essential services
- Route safety and historical incident patterns
- Weather and road closures
- Worker's shift timing
- Emergency facilities
- Real-time changes and unreliable/incomplete data

Given "I finish work at 2:15 AM and need to reach home by 3:00 AM," the system should generate and continuously update the safest feasible plan. The system must also keep working when live data becomes unavailable, and it should distinguish KNOWN → PROBABLE → OUTDATED → UNKNOWN rather than confidently inventing information.

THEME 1 • PROBLEM 2

THE INVISIBLE WORKER

CONTEXT

A street vendor may work 10–12 hours every day but still has no reliable understanding of:

- True accrual profit (vs. daily cash on hand)
- Latent inventory and dynamic portion sizes
- Implicit customer credit and aliases
- Unrecorded spoilage vs. theft vs. personal consumption
- True cost of goods sold (COGS) without a formal Bill of Materials

Most accounting software assumes formal businesses, reliable cloud connectivity, explicit touchscreen data entry, and clean inputs.

CHALLENGE

Design a fully offline, on-device probabilistic ledger engine that turns noisy multi-channel evidence into a mathematically sound business picture. The system must passively accept chaotic, contradictory inputs, such as a crumpled, oil-stained wholesale slip combined with ambient audio ("Bhai 3 kilo tamatar liya 180 mein... [85dB traffic noise] ...Sharma ji ka 40 jama kar, aur 7 plate poha de."), and reconstruct a double-entry ledger from chaos, outputting ranges and confidence intervals rather than blind point guesses. It must autonomously handle:

- **Zero-shot recipe inference:** Deducing unknown ingredients per plate from raw material burn rates alone, tracking dynamic portion sizes without a Bill of Materials.
- **Loss attribution:** Splitting unrecorded stock loss into spoilage, theft, and personal consumption as a probability distribution based on environmental covariates.
- **Cross-modal fusion:** Merging speech, delayed SMS chimes, and handwritten tallies into a single event without triple-counting or dropping partial evidence.
- **Dynamic linguistics:** Adapting to hyper-local code-switching and non-standard, fluctuating units via a mid-event Lexicon Drop.

The system must also run across three devices shared by the family. When devices briefly reconnect via Bluetooth, it must execute Conflict-Free Replicated Data Type (CRDT) sync to merge conflicting ledgers, and it must survive hostile clocks (devices randomly resetting to 1970 or jumping forward) without reordering or overwriting history, resolving contradictory credit claims while maintaining perfect double-entry balance.

PROBABILISTIC EVALUATION CHAIN



THEME 1 • PROBLEM 3

THE HUMAN-POWERED EMERGENCY NETWORK

CONTEXT

When a city floods and cell towers go dark, thousands of capable citizens are stranded within kilometers of people who urgently need help. Without a central server, there is no reliable understanding of:

- True availability and physical capability of responders (vs. stale claims)
- Trustworthiness of distress calls (real emergencies vs. decoys)
- Dynamic hazard zones and safe routing
- Actual battery life and device-to-device network topology
- Triage urgency across competing, unverified victims

Most emergency software assumes centralized dispatch, active internet, verifiable identities, and trusted GPS.

CHALLENGE

Design a fully decentralized, on-device, multi-agent mesh protocol that safely matches victims to responders using only peer-to-peer radio links in an active, adversarial disaster zone. The system must passively accept fragmented, untrusted, strictly constrained inputs, such as “Need insulin for dad, ground floor flooded” (broadcast over mesh, received with drifting GPS coordinates and 12% battery), and orchestrate safe, confirmed physical dispatches without a central authority. It must autonomously handle:

- **Adversarial resilience:** Defending against Sybil attacks (one node pretending to be 40 nurses), location spoofing, and malicious decoy emergencies.
- **Zero-shot capability matching:** Parsing open-vocabulary, code-mixed claims (“has a boat”, “can do dialysis”) for completely novel disaster types.
- **Privacy under surveillance:** Routing sensitive health/location data through untrusted relay nodes without exposing victims to passive eavesdroppers.
- **Distributed triage and opportunity cost:** Overcoming the bystander effect and ensuring the only qualified medic isn’t sent to a minor emergency while a critical one is ignored.
- **Human-in-the-loop acceptance:** Generating one-line, plain-language justifications that convince simulated human responders to accept the physical risk of moving.

The network will also undergo severe physical partitions where disconnected “islands” of devices independently resolve the same emergencies. When the mesh reconnects, agents must autonomously resolve conflicts, cleaning up double-dispatches and orphaned victims without canceling a responder already at the door, and survive the “Red Team Hour,” where rival teams inject attacker nodes designed to forge medical credentials, drain network battery, and lure responders into rising floodwaters.

PROBABILISTIC EVALUATION CHAIN

Claim Verification → Hazard Avoidance → Trust Scoring → Battery/Routing Cost → Safe Dispatch

THEME 2

CRISIS BEFORE CRISIS

Core idea: Predict the conditions that create disasters instead of only responding after they happen.

THEME 2 · PROBLEM 1

THE FIRST 10 MINUTES

CONTEXT

A disaster rarely begins at the moment it becomes visible. It is a Tuesday evening in a city of eight million people. At 00:00 the rain gets heavier. At 00:03 drainage sensors report anomalies. At 00:05 traffic slows. At 00:07 emergency lines get busy. At 00:09 two hospitals report unusual arrivals. The duty officer has one question and one button: is this noise, a broken sensor, a minor local incident, or the first ten minutes of a cascading crisis? Waiting costs lives, but pressing the button for nothing costs credibility and trains the city to ignore the next real alarm.

CHALLENGE

Build a continuous, online prediction and decision system that ingests a live, messy, multi-source event stream and outputs a calibrated probability of a crisis, a recommended action, and a forecast of what breaks next, updating its belief every minute. The stream is irregular, asynchronous, heterogeneous, and unreliable:

- Sensors with unknown, changing reliability and correlated failures (e.g., one gateway outage masking 100 sensors)
- Dispatch logs and delayed citizen reports
- Retractions, duplicates, and out-of-order events
- Coordinated rumor waves acting as mimics

The system must recommend an action under a strict cost model: WATCH (keep monitoring) · PREPARE (cheap readiness) · ALERT (notify agencies) · ESCALATE (public alert). The important part is inferring the true hidden state from the first faint signals. The stream might look like a cascade but could be explained by heavy rain (drainage and traffic), normal storm emergency calls, a flu cluster at the hospitals, and a gateway outage.

> **Core Challenge:** Be calibrated when it cannot know for sure, commit fast when the evidence solidifies, and explain exactly why it changed its mind.

IT MUST ALSO MEET THESE REQUIREMENTS

- **Honest Probabilities & The Twin Test:** Maintain calibrated probabilities over four hypotheses (noise, sensor failure, isolated incident, cascading crisis). The system faces identical 7-minute prefixes that diverge, and is penalized for confident wrong answers before the split and rewarded for fast commitment after.
- **Optimal Stopping & Cost Model:** Weigh the value of lead time (damage avoided) against the penalty of false alarms (city fatigue/cry-wolf effect).
- **Foresight:** Forecast the next three events with time windows, proving it understands the causal graph and not just correlation.
- **Learning & Prior Shifts:** Autonomously update trust in data sources across episodes and adapt to silent base-rate shifts (e.g., monsoon season starts) without offline retraining.
- **Adversity & Cutouts:** Handle stream cutouts (e.g., all cameras vanish mid-episode) and structured lies (common-cause failures) without crashing or hallucinating.
- **Stability:** Avoid alert flip-flopping (ESCALATE → WATCH → ESCALATE), which erodes trust.

THEME 2 • PROBLEM 2

THE 72-HOUR WARNING

CONTEXT

A disaster rarely begins at the moment it becomes visible. A flood may begin with rainfall, drainage saturation, rising water levels, blocked outlets, power failures, traffic congestion, and delayed evacuations. A hospital crisis may begin with rising patient arrivals, staff shortages, ambulance delays, and medicine consumption days earlier.

CHALLENGE

Build a system that predicts cascading urban crises up to 72 hours before they become critical, discovers how city systems depend on each other, and continuously revises its forecasts as evidence arrives. The system receives asynchronous, heterogeneous, unreliable streams:

- Weather, rainfall, and river/water levels
- Traffic and public transportation
- Power availability and infrastructure failures
- Hospital capacity, ambulance movement, and emergency calls
- Satellite and geospatial data
- Historical incidents and citizen reports

It must produce four risk states, per zone and city-wide: NORMAL · WATCH · HIGH RISK · CRITICAL. The important part is inferring how an initial disruption propagates across interconnected urban systems, identifying the most probable failure sequence, and estimating when each stage will occur and when the situation becomes critical. For example: Heavy rainfall → Drainage capacity decreasing → Traffic and road accessibility deteriorating → Emergency response delays increasing → Hospital capacity approaching its limit → Power substation exposure and communication load rising → CRITICAL CASCADE.

Core Challenge: Predict the cascade, not merely the individual event.

IT MUST ALSO MEET THESE REQUIREMENTS

- **Multi-Horizon Forecasting:** Predict risk from the next hour to 72 hours ahead, with calibrated probabilities and time-to-critical estimates, not just labels.
- **Dependency Discovery:** The interdependency structure between urban systems is not provided. Infer it from data and update it as the city changes.
- **Cascade Prediction:** Forecast stage-by-stage propagation, including branching, feedback loops, and simultaneous cascades that interact.
- **Continuous Online Learning:** Update beliefs on streaming data under concept drift, without retraining from scratch or waiting for a complete dataset.
- **Uncertainty and Adversity:** Handle missing, delayed, noisy, contradictory, and deliberately misleading inputs (sensor failures, false citizen reports), with confidence that degrades gracefully.
- **Causal Explainability:** Identify the causal drivers of every risk change and present the predicted chain of failures with evidence behind each link.
- **Constrained Counterfactual Intervention:** Recommend preventive actions under limited resources, estimate how each would break the cascade, and account for side-effects (e.g., evacuation increasing congestion).
- **Generalization:** Perform on a city or scenario not seen during development.
- **Temporal Validation:** Demonstrate lead time, false-alarm rate, and calibration on replayed streams with strict future-data isolation.

THEME 2 • PROBLEM 3

THE CRITICAL 1%: HIDDEN COMMUNITY VULNERABILITY

CONTEXT

City planners and emergency responders operate on outdated census data and static maps, with no reliable, real-time understanding of:

- Dynamic infrastructure decay (water pressure drops, rolling blackouts)
- Hidden transit deserts and instantly blocked evacuation routes
- Micro-economic constraints preventing rapid community evacuation
- Healthcare deserts combined with localized warning-system dead zones
- Cascading historical failures that compound neighborhood risk

Most vulnerability mapping assumes static populations, reliable annual census updates, and homogeneous neighborhoods, all of which fail during an actual crisis.

CHALLENGE

Design an autonomous, privacy-preserving spatial intelligence engine that identifies hyper-local, systemic vulnerability clusters and root causes without tracking, storing, or inferring any individual's personal data. The system must passively accept fragmented, noisy, unstructured urban "exhaust," such as "Bus route 4 suspended, substation C voltage dropping, and 300 anonymized localized emergency call drops over 48 hours," and reconstruct a causal, geographic vulnerability map. The output should be "This area has unusually high crisis vulnerability," not "This specific person is vulnerable." It must autonomously handle:

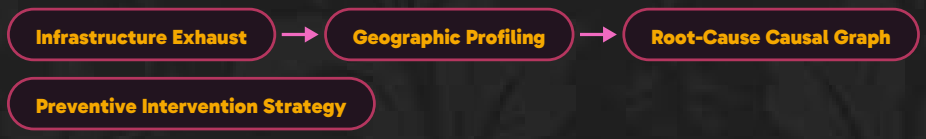
- **Strict privacy preservation:** Extracting structural vulnerability (e.g., "this block has no backup power and high medical need") without exposing or retaining PII. It must guarantee zero PII leakage and survive a simulated "Red Team" re-identification attack.
- **Cross-domain data fusion:** Synthesizing unstructured transit logs, intermittent IoT grid sensors, economic proxies, and sparse public health records.
- **Zero-shot crisis prediction:** Identifying novel, unprogrammed vulnerability patterns (e.g., a temporary construction site trapping a low-income housing block).
- **Explainable causality:** Outputting clear, actionable logic (e.g., "High risk due to 80% grid failure history + >2km to nearest clinic") and avoiding black-box scoring. The causal graphs must be clear enough to be validated by simulated urban planners.

The system must also operate during a simulated cascading infrastructure failure where government data contradicts crowdsourced civilian reports and 60% of input data streams are abruptly severed. It must dynamically recalculate geographic risk in real time, relying entirely on decentralized federated learning across isolated municipal nodes to maintain strict privacy under adversarial conditions. Accuracy is measured by spatial overlap (Intersection over Union) with hidden ground-truth vulnerability zones.

CONCRETE DELIVERABLES

- **Spatial Inference Engine:** A backend service that processes live, messy data into privacy-preserved risk clusters.
- **Dynamic Causal Graph:** An explainability dashboard showing why a specific zone is vulnerable (the converging variables).
- **Automated Intervention Log:** Generated recommendations (e.g., "Deploy mobile generators to Sector 4; dispatch localized transit to Sector 7").

PROBABILISTIC EVALUATION CHAIN



THEME 3

ATTENTION ECONOMY

Core idea: Build technology that protects human attention instead of extracting it.

THEME 3 • PROBLEM 1

THE ATTENTION FIREWALL

CONTEXT

Modern applications compete for attention through notifications, recommendation algorithms, infinite feeds, and engagement loops. Existing digital wellbeing tools mostly measure usage after it happens.

CHALLENGE

Develop a mobile application that acts as an intelligent, real-time attention firewall, identifying and intercepting potentially manipulative digital interactions before they disrupt the user's intended activity.

Core Features:

- Detect notification bursts and repetitive interruption patterns
- Classify interruptions by urgency, relevance, and user-defined priorities
- Build a smart notification queue that releases notifications at suitable moments
- Learn from user decisions without sharing sensitive behavioral data

Distinguishing genuinely urgent messages from engagement-driven interruptions while preserving user autonomy.

Advanced: On-device ML, notification scheduling, contextual classification, privacy-preserving personalization.

THEME 3 • PROBLEM 2

THE INFINITE SCROLL INTERVENTION ENGINE

CONTEXT

Infinite feeds and autoplay can turn a short intended session into prolonged, unplanned consumption.

CHALLENGE

Build a mobile app that detects when actual usage diverges from a user's original intention and introduces personalized interventions without simply blocking access.

Core Features:

- Ask users to define their purpose and intended duration before a session
- Detect session overruns and repeated attempts to exit
- Offer context-aware stopping points and exit prompts
- Distinguish intentional long-form consumption from compulsive browsing
- Measure whether interventions improve user-defined outcomes

Detecting loss of intentionality without assuming that all long screen time is harmful.

Advanced: Usage pattern analysis, Android usage-access APIs, intervention experiments, behavioral modeling.

THEME 3 • PROBLEM 3

THE ATTENTION DIGITAL TWIN

CONTEXT

Everyone's attention patterns are different.

CHALLENGE

Build a personal attention digital twin that learns:

- When the user focuses best
- What causes interruptions
- What types of content cause prolonged sessions
- Which notifications actually matter
- When the user is mentally fatigued
- Which applications fragment concentration

The system should simulate: "If I allow this notification right now, what is likely to happen?" Example: The user is coding and receives "Someone liked your post." Predicted outcome: 2 min interruption → social app opened → 17 min browsing → task resumption delay of 8 min. The system can therefore suppress the interruption. Predictions must also be counterfactual: what would probably have happened if the notification had been shown?

THEME 4

INFORMATION ASYMMETRY

Core idea: Give ordinary people the information that institutions already have, and make it impossible for that information to be hidden again.

THEME 4 • PROBLEM 1

THE GOVERNANCE ILLUSION: DAO ACCOUNTABILITY

CONTEXT

DAOs promise transparent community voting, but core contributors and delegates hold hidden, asymmetric information. Current systems assume transparent on-chain voting equals informed consent, but critical data (financial implications, technical risks, conflicts of interest, implementation details) is scattered across disconnected forums, PDFs, and social media.

CHALLENGE

Design a decentralized governance engine that connects proposals, supporting evidence, voting, and execution in one verifiable system, preventing central authorities from controlling the information available to ordinary token holders. The system must passively accept fragmented, obfuscated governance data, such as "Forum post proposing a $5M treasury grant, linked to a vague external PDF, with hidden wallet ties to the proposer," and reconstruct a cryptographically binding, auditable decision trail. It must autonomously handle:

- **Verifiable proposals:** Recording proposal versions, financial assumptions, and technical assessments using on-chain commitments and tamper-evident histories.
- **Decision intelligence:** Generating understandable summaries of treasury impacts and dependencies with traceable references to the underlying evidence.
- **Conflict-of-interest disclosures:** Forcing proposers to publish verifiable disclosures, mathematically mapping wallet connections.
- **Execution accountability:** Automatically comparing approved human-readable decisions against actual on-chain smart contract execution, flagging material discrepancies.

The system must also operate during a live, contested protocol upgrade in which a proposer deliberately obfuscates a malicious smart contract payload in technical jargon. It must parse the intended human-readable proposal, simulate execution of the actual submitted bytecode, and autonomously halt or flag the transaction if the code deviates from what the community voted for.

PROBABILISTIC EVALUATION CHAIN



THEME 4 • PROBLEM 2

THE BLACK HOLE LEDGER: VERIFIABLE NGO FUND FLOWS

CONTEXT

Donors send millions to NGOs but have no independently verifiable way to track whether contributions reached their intended destination. Current systems assume financial reports and self-published impact metrics are truthful, leaving donors dependent on the organization's own claims. Transactions, expense records, and impact reports are completely disconnected.

CHALLENGE

Design a decentralized donation-tracking system that establishes an auditable connection between donated funds, authorized spending, and supporting evidence, without exposing beneficiaries' sensitive information. The system must passively accept complex, multi-hop financial flows, such as "A 50 ETH donation earmarked for 'water infrastructure' is swapped, bridged, and supposedly spent off-chain via local fiat vendors," and reconstruct a traceable, verifiable timeline of impact. It must autonomously handle:

- **Purpose-bound allocation:** Using smart contracts to manage earmarked funds and enforce agreed disbursement conditions.
- **Verifiable expenditure records:** Linking outgoing on-chain transactions to off-chain invoices/receipts using cryptographic commitments.
- **Privacy protection:** Shielding beneficiaries' identities and sensitive documents from the public blockchain while maintaining mathematical proof of delivery.
- **Discrepancy detection:** Flagging unexplained transfers, spending outside declared categories, or inconsistencies between published reports and recorded transactions.

The system must also track funds across multiple chains and off-chain fiat off-ramps while an adversarial NGO tries to double-count a single piece of off-chain expenditure evidence across multiple donor reports and deliberately forges 30% of its receipts. It must detect the cryptographic duplication and freeze escrow disbursements without relying on a centralized auditor.

PROBABILISTIC EVALUATION CHAIN

Inbound Donation → Escrow Condition → Off-Chain Invoice Proof

Cryptographic Impact Verification

THEME 4 • PROBLEM 3

THE IDENTITY MONOPOLY: DECENTRALIZED PORTABLE PROOFS

CONTEXT

Institutions monopolize the evidence of an individual's achievements (degrees, certifications, work experience). Current systems assume users must repeatedly ask centralized databases to prove their own qualifications, often over-sharing sensitive data (e.g., exposing a full transcript just to prove a degree exists).

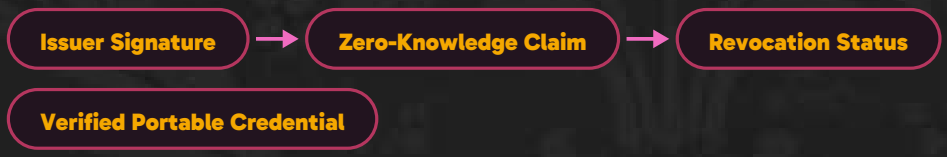
CHALLENGE

Design a user-controlled credential verification engine that lets individuals hold portable, cryptographically verifiable proofs and selectively disclose claims using zero-knowledge technology. The system must passively accept complex, multi-party credential interactions, such as "An employer demands proof of a specific university degree, age > 21, and an active medical license, without knowing the user's date of birth, name, or home address," and construct a mathematically sound verification layer. It must autonomously handle:

- **Issuer-signed credentials:** Letting third parties check issuer signatures, credential integrity, and issuer status without contacting the institution.
- **Selective disclosure & ZK proofs:** Establishing claims (such as meeting an age threshold or holding a qualifying credential) without disclosing the underlying raw data.
- **Revocation registries:** Letting issuers publish verifiable credential status updates (including revocation) without exposing a public history of an individual's credentials.

The system must also survive a simulated breach in which a trusted major university's issuance key is compromised and malicious actors flood the network with fake degrees. It must dynamically update decentralized trust registries and mathematically isolate fraudulently issued credentials from the last 48 hours without invalidating legitimate legacy degrees, while maintaining absolute zero-knowledge privacy for all innocent users.

PROBABILISTIC EVALUATION CHAIN



RULES & GUIDELINES

- Teams must consist of 3–4 members with accurate registration details.
- All submissions must be original or appropriately credited; plagiarism is prohibited.
- Multiple theme participation and prize stacking are permitted, subject to eligibility.
- Self-voting in the Stars system is prohibited.
- Participants must follow organiser instructions and maintain respectful conduct.
- Judges' decisions will be final, subject to the official event rules.

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